The Asymmetric Spillover Effects of Underlying Asset Volatility on NFT Liquidity

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Market Phenomenon and Hypothesis

- **Market Friction**: NFT liquidity relies heavily on underlying assets (e.g., ETH). During macroeconomic turbulence, NFTs frequently face severe liquidity dry-ups.
- **Flight to Quality Hypothesis**: The dynamic contagion induced by macro volatility is **asymmetric**. Capital panics and withdraws from tail NFT projects while concentrating on blue-chip NFTs as a relative safe haven.

Research Objective

To quantify this asymmetric spillover mechanism while explicitly addressing market microstructure noise, jump components, and endogeneity/identification issues inherent in crypto-asset markets.

Stylized Facts: Macro Shocks and Market Freezes

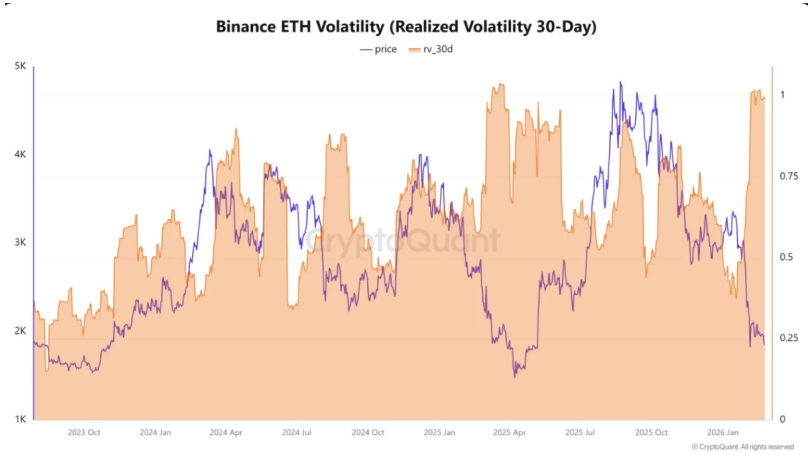

Figure 1: Spikes in ETH Volatility

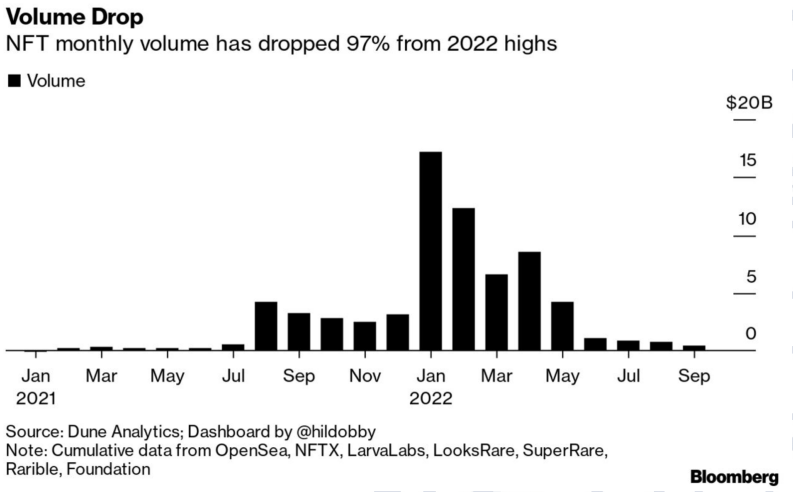

Figure 2: Collapse in NFT Volume

- **Empirical Observation**: Extreme spikes in ETH volatility directly coincide with a sudden drying up of bids in the NFT secondary market.

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NFT Panel Construction and Sparsity

To mitigate selection bias and estimation errors from extreme data sparsity, we employ a Panel Fixed Effects structure:

- **Objective Stratification**:
  - *Blue-chip*: Top 5% of collections ranked by historical 6-month daily volume.
  - *Tail*: Bottom 50% (excluding inactive projects with zero trades over 6 months).
- **Robust Treatment of Zero-trading Days**:
  - *Volume*: True zeros are retained as informative signals via $\ln(1 + Volume)$.
  - *Floor Price*: To avoid artificially inflating liquidity via simple LOCF, we deploy a triple-check strategy: (A) Drop NaNs, (B) standard LOCF, and (C) Missing Indicator Dummy Variable in panel regressions.

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Measure Construction: Jumps vs. Continuous Volatility

Addressing Microstructure Noise and Jumps

We sample at 1-min, 5-min, and 15-min frequencies to test estimator stability. To separate continuous volatility from discrete jumps (Barndorff-Nielsen & Shephard, 2004), we calculate:

$$RV_t = \sum_{i=1}^{n} r_{t,i}^2 \tag{1}$$

$$BPV_t = \frac{\pi}{2} \sum_{i=2}^{n} |r_{t,i}||r_{t,i-1}| \tag{2}$$

The jump component $J_t = \max(RV_t - BPV_t, 0)$ and continuous component BPV_t will be independently regressed against NFT liquidity to test their distinct impacts.

Dynamic Identification: VAR vs. Local Projections

VAR with Economic Cholesky Ordering

We impose a strict causal ordering: $MacroVIX \rightarrow ETHVolatility \rightarrow BlueChip \rightarrow Tail$. *Economic Justification*: Macro VIX represents exogenous global risk; highly liquid ETH prices macro-news instantaneously. Conversely, NFT markets are thin and exhibit slower information transmission compared with highly liquid ETH markets.

Identification Robustness

1. **Local Projections (LP)**: Concurrently estimate IRFs using Jordà (2005) to mitigate VAR misspecification risks.
2. **Natural Experiments**: Introduce quasi-exogenous shocks (e.g., FTX crash, USDC depeg) analyzing formal event windows of $[-3, +3]$ or $[-7, +7]$ to strengthen causal claims.

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Expected Academic and Practical Contributions

- **Contribution 1: Novelty in Spillovers**
  First study to analyze asymmetric spillovers from underlying crypto volatility to NFT liquidity, bridging macro-crypto shocks with micro-NFT market structure.

- **Contribution 2: Methodological Rigor**
  Decomposes high-frequency volatility into jump versus continuous components, isolating the exact driver of liquidity dry-ups.

- **Contribution 3: Market Dynamics**
  Provides robust empirical evidence of *flight-to-quality* dynamics in NFT markets, challenging the assumption that all digital collectibles share homogeneous risk exposures.

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Selected References I

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Thanks!

Q & A